

System Identification II: Observer/Kalman Filter Identification (OKID)

Identifying lightly damped systems – and shrugging off the unknown initial state – with an embedded observer.

The plain impulse-response fit of `Intro.m` needs about one Markov parameter per sample of settling time, which is hopeless for lightly damped or oscillatory systems. **OKID** (Juang) sidesteps this: it identifies the Markov parameters of an **observer** form whose response dies out in a handful of steps (a deadbeat observer), then algebraically recovers the true system Markov parameters.

This tutorial drives the repository's own root-level helpers – `YV_Form_nonzero`, `pinv2`, and `recover_SYSP` – the same tools behind the `Final/` exam problems. You will:

- form the observer regression and solve it with a robust pseudoinverse,
- recover the **observer** then the **system** Markov parameters, and
- confirm they match the true impulse response, even from a nonzero initial state.

Run with `publish('OKID.m')`, or step through with **Ctrl+Enter**.

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A known (undamped) system

An undamped oscillator sampled at $T = 0.2$ s – the kind of ringing system a plain FIR fit chokes on, because its impulse response never decays.

```
A_c = [0 1; -1 0];
B_c = [0; 1];
C = [1 0];
D = 0;
T = 0.2;
sysd = c2d(ss(A_c,B_c,C,D), T);
A = sysd.A; B = sysd.B; Cd = sysd.C; Dd = sysd.D;
n = size(A,1);
q = size(Cd,1);           % outputs
m = size(B,2);           % inputs
```

Run an experiment (with a nonzero initial state)

Random input, and – to make the point – a nonzero initial condition the identifier is never told about.

```
rng(1);
ll = 60;
u = randn(1,ll);
x0 = [1; 0];
Y = lsim(sysd, u(:), (0:ll-1)*T, x0).';    % Y is q-by-ll
```

The deadbeat observer at the heart of OKID

Pick an observer gain G that places the eigenvalues of $A + GC$ at the origin. Then $\bar{A} = A + GC$ is **nilpotent**, so only $p = n$ observer Markov parameters are nonzero and the identification is finite.

```
G = acker(A', -Cd', zeros(1,n)).';
p = n;
fprintf('eig(A+GC) = %s    (deadbeat: all ~0)\n', mat2str(eig(A+G*Cd).',3))

eig(A+GC) = [1.48e-16+1.94e-08i 1.48e-16-1.94e-08i]    (deadbeat: all ~0)
```

Solve the observer regression with the repo helpers

`YV_Form_nonzero` stacks the input/output record into the regression matrices $\bar{Y}, \bar{V}$; `pinv2` is a tolerance-truncated SVD pseudoinverse that keeps the solve well conditioned. Their product is the row of observer Markov parameters.

```
[Y_bar, V_bar] = YV_Form_nonzero(u, Y, p);
cap_y_hat = Y_bar.' * pinv2(V_bar, 1e-5);
```

Recover the system Markov parameters

`recover_SYSMP` runs the OKID recursion that converts the **observer** Markov parameters into the **system** Markov parameters $Y_k = CA^{k-1}B$.

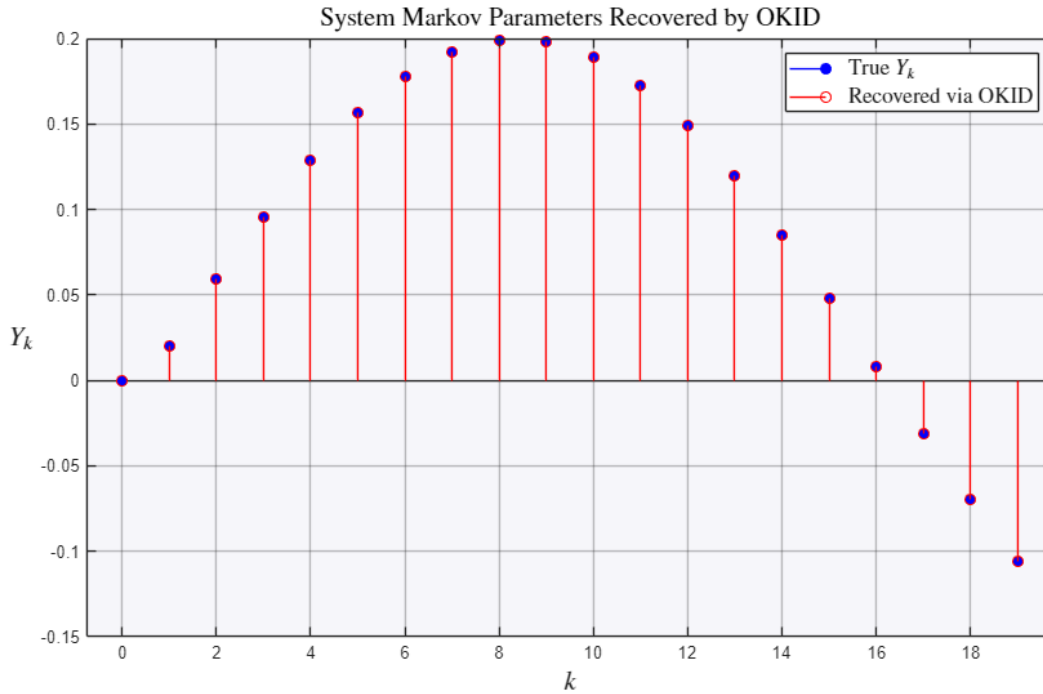
```
n_mp = 20;
RSMP = recover_SYSMP(cap_y_hat, n_mp, q, m);

SMP = zeros(q, n_mp);          % true values for comparison
SMP(:,1) = Dd;
for i = 2:n_mp
    SMP(:,i) = Cd*A^(i-2)*B;
end

figure
stem(0:n_mp-1, SMP, 'b', 'filled')
hold on
stem(0:n_mp-1, RSMP, 'r')
hold off
grid on
legend('True  $Y_k$ ', 'Recovered via OKID', 'Interpreter', 'latex', 'FontSize', 13)
title('System Markov Parameters Recovered by OKID', 'Interpreter', 'latex', 'FontSize', 15)
ylabel('Y_k', 'Interpreter', 'latex', 'FontSize', 16); set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('k', 'Interpreter', 'latex', 'FontSize', 16)

fprintf('Relative error, recovered vs. true Markov parameters: %.2e\n', ...
    norm(RSMP - SMP)/norm(SMP))
```

```
Relative error, recovered vs. true Markov parameters: 1.73e-14
```



Why it worked from a nonzero initial state

Notice that we never handed the identifier x_0 . The `"_nonzero"` in `YV_Form_nonzero` refers to exactly this: the regression starts only after the deadbeat observer has forgotten the initial condition ($k \geq p$), so the unknown x_0 drops out of the equations.

Try it yourself

- Add measurement noise – change the output line to `Y = lsim(...).' + 0.005*randn(q,11);` – and watch `pinv2`'s tolerance keep the recovery stable (try loosening it to `1e-2`).
- Increase the record length `11` and see the recovery error shrink.
- Swap in a lightly damped plant, `c2d(ss(tf(1,[1 0.2 1]),T),T)`, and confirm OKID still needs only `p = n` observer parameters.

Next: `ERA.m` turns these Markov parameters into a state-space model (A, B, C) .

Summary

- OKID identifies the Markov parameters of a deadbeat **observer** form, which are finite in number even for undamped systems.
- The root helpers `YV_Form_nonzero` / `pinv2` / `recover_SYSP` do the regression and the recursion back to system Markov parameters.
- Starting the regression after the observer settles makes the method immune to the unknown initial state.